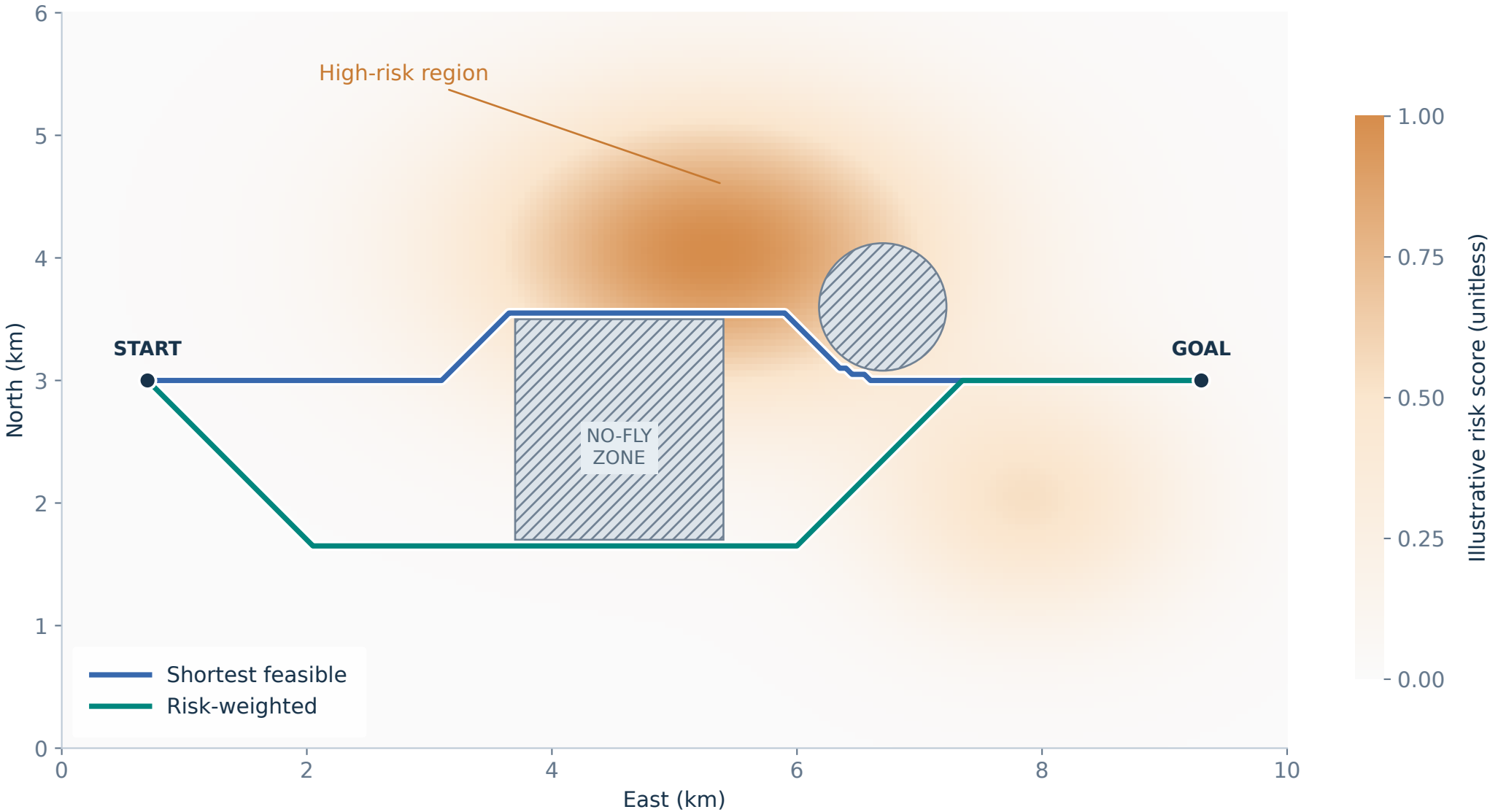
01 / UAV PATH PLANNING

A shorter route can carry more exposure



SHORTEST FEASIBLE

9.06 km | exposure 3.24 score-km

RISK-WEIGHTED

9.72 km | exposure 1.01 score-km

Synthetic mathematical example. Same 50 m grid, obstacles and endpoints; Dijkstra weights 1 or (1 + 3r).
Risk is an invented scalar field. Distances and exposure are grid-path values, not aircraft performance.